MATH 11 — ELEMENTARY STATISTICS

§1.1

What is Statistics?

The science of data, and the two things we do with it

Today’s Goal

After completing this section, you should be able to:

- Explain what statistics is and why it is important
- Distinguish between descriptive and inferential statistics
- Identify real-world applications of statistical methods

Would You Believe It?

A headline says a new drug is 95% effective. What would you want to know before you believed it?

Who was studied, how many of them, how the number was worked out, and whether it holds for people who were never in the study. Statistics is the field that answers those questions.

The Science of Data

At its core, **statistics** is the science of collecting, organizing, analyzing, and interpreting data to make informed decisions.

- **Make better decisions.** Weigh evidence and risk — a treatment, an investment, a claim in the news.
- **Understand research.** Medicine, psychology, and economics all report their conclusions statistically.
- **Detect misleading information.** Statistics can be misused to deceive. Knowing how it works is how you catch that.

Statistics at Work

- **Medical research.** A hospital network tries a new asthma inhaler with 12,000 patients; emergency visits drop 38%.
- **Quality control.** A bottling plant checks 300 bottles, finds 4 underfilled, and estimates the rate for the whole run.
- **Sports analytics.** A basketball team models which shots to take and which players to pair on the floor.
- **Public policy.** A city uses census and survey data to decide where to add bus routes.

Same Data, Two Claims

Which claim goes beyond the 240 students we counted?

Claim A. Of the 240 students in our survey, 150 have declared a major.

Claim B. Based on those 240 students, about 6 in 10 Fresno State students have declared a major.

Claim B. It describes students nobody looked at. That leap is the difference between the two branches of statistics.

Staying Inside the Data

Definition 1.1.2

Descriptive statistics refers to the methods used for collecting, organizing, summarizing, and presenting data.

Tables, graphs, averages, percentages — anything that describes the group in front of you and stops there.

Descriptive statistics is the focus of **Chapters 1–3**.

Describing What We Measured

A university reports that the average GPA of its graduating class is 3.06, with 38% of graduates earning honors.

These are descriptive statistics. They summarize the data for this particular group of graduates.

Notice what is *not* claimed: nothing about next year's class, and nothing about any other university.

Reaching Past the Data

Definition 1.1.4

Inferential statistics refers to the methods used for drawing conclusions about a population based on data from a sample, using probability to assess how confident we can be in those conclusions.

We do this because we almost never get to measure everyone. A sample is what we can afford.

Inferential statistics is the focus of **Chapters 4–12**.

Reaching Beyond the Sample

A polling organization surveys 1,200 likely voters and finds that 54% support Candidate B.

They conclude that Candidate B leads among *all* likely voters, with a margin of error of $\pm 3\%$.

That claim covers millions of people the pollsters never spoke to. The margin of error is how they admit the uncertainty: somewhere between 51% and 57% — still a lead either way.

Now You Try: Which Branch?

You Try 1.1.1. Is each statement descriptive or inferential?

1. Of the 45 games the team played, it won 28. **Descriptive.**
2. A taste test of 80 shoppers suggests customers statewide prefer the new recipe. **Inferential.**
3. From 500 households, the agency estimates county median income is $71,000. **Inferential.**

The test: does the claim stay inside the group measured, or step outside it?

Why Probability Is Here

Probability is the mathematical foundation that connects descriptive and inferential statistics. When we use sample data to make claims about a population, we can never be 100% certain our conclusions are correct. Probability lets us measure that uncertainty and say how confident we are.

Definition 1.1.6

The **probability** of an event is a number between 0 and 1 (inclusive) that represents how likely it is that the event will occur.

Is This Coin Fair?

Every NFL game starts with a coin toss. We assume the coin is fair — tossed many times, heads about half the time.

Suppose we flip one particular coin 60 times and get 44 heads and 16 tails.

A fair coin would give about 30 heads. We got 44.

Probability asks: *if this coin really were fair, how often would it do something this extreme?* If almost never, we doubt the coin.

That reasoning is **hypothesis testing** — **Chapters 9–11**.

Statistics Is Not Mainly Arithmetic

Warning 1.1.8

As we progress through this course, you will be presented with many formulas and procedures. This may mislead you into thinking that statistics is mainly about calculating things. This could not be further from the truth.

What Your Job Actually Is

Warning 1.1.8, continued

While calculations are necessary, it is more important that you understand **how to interpret the results** of these calculations and procedures. Modern technology can perform statistical calculations quickly and accurately. Your job is to understand what the numbers mean and how to use them to make informed decisions.